

General Requirements

1. SCOPE

1.1 Intent. This standard specification invokes general requirements for conducting vessel repairs performed by commercial contractors at Coast Guard Base Honolulu aboard Coast Guard Cutter Kimball.

2. REFERENCES

COAST GUARD DRAWINGS

None

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General Requirements

Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2022, Welding and Allied Processes

Surface Forces Logistics Center Standard Specification 5550 (SFLC Std Spec 5550), 2022, Fire Prevention and Response

OTHER REFERENCES

None

3. REQUIREMENTS

3.1 General. The Contractor must conform to all requirements specified in SFLC Std Spec 0000 and in this item, as applicable, during the performance of this availability. The requirements of this WI applies to all work under the scope of this contract, whether explicitly stated in all following work items or not, and to all other work subsequently authorized by changes, modifications, or extensions to the contract.

3.2 Contractor-provided fire watch personnel. The Contractor must provide fire watch personnel and equipment.

3.3 Welding and brazing requirements. The Contractor must perform all welding and allied processes, and NDE in accordance with SFLC Std Spec 0740.

3.4 Fire Prevention Specification. The Contractor must submit a Fire Prevention Plan in accordance with paragraph 3.1 of SFLC Standard Spec 5550

3.5 Period of Performance. 16FEB2026 to 06MAR2026

3.6 Location. 400 Sand Island Pkwy, Honolulu, HI 96819

3.6 Point of Contact. Contact LT Parker Brunkala via email at Parker.T.Brunkala@uscg.mil with any questions regarding this scope of work.

3.7 Base Access. All personnel who enter Base Honolulu must be enrolled in RAPID Gate. Contractors must enroll using the documentation sent separately from this scope of work. Contractor must adhere to Kimball's work hours unless given written approval from the KO.

3.8 Selection Criteria. All proposals will be evaluated based on thoroughness, cost, and past performance.

WORK ITEM 1: Air Conditioning Plant, Repair

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to renew the AC condenser and compressor for the #1B circuit, flush the refrigerant side of the #1B circuit, renew or overhaul various components on the refrigerant side of the #1B circuit and restore the plants back to service on the USCG Cutter Kimball in Honolulu, HI. Location of work is Air Conditioning Equipment Room, Compartment 3-22-0-Q. The USCG has determined that the #1B refrigerant circuit was exposed to seawater.

1.2 Government-furnished property.

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
Y	Condenser, 130 Ton Chill Water	NSN: 4130-01-549-7737	1 ea.	54,287
Y	York Semi-hermetic screw compressor	NSN: 4130-01-549-7925	1 ea	69,474.00
N	Indicator, Moisture	NSN: 4130-01-670-3068	1 ea	21.89
N	Valve, Thermal Expansion, Economizer	NSN: 4820-01-583-7241	1 ea	1037.89

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 750 WMSL 508-001, Rev C, Piping and Machinery Insulation Schedule and L/M

Coast Guard Drawing 750 WMSL 516-001, Rev E, A/C & Refrigeration Piping Diagram

Coast Guard Drawing 750-WMSL_520_001, Rev H, Seawater Cooling System Diagram (ASC520001)

Coast Guard Drawing 750-WMSL-520-004, Rev E, Salt Water Cooling Unit 1320 (ASC520132-SW)

Coast Guard Drawing 750-WMSL_520_506, Rev E, Auxiliary Seawater Piping Mods FWD of FR 52.

Coast Guard Drawing FL-1204-3, Rev A, Fleet Welded Padeyes and Links

COAST GUARD PUBLICATIONS

Coast Guard Technical Publication (TP) 7608, SWBS 514, February 2012, York/Thermaflo Oz Saver Portable Refrigerant Recovery Unit, Model 4000 (418 WMSL)

Coast Guard Technical Publication (TP) S9514-F1-MMA-010, SWBS 514, Jan 2012, Air Cond Plant HFC-134A 130 Ton Capacity Description, Operation and Maintenance Manual

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General Requirements

Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2022, Welding and Allied Processes

Surface Forces Logistics Center Standard Specification 5000 (SFLC Std Spec 5000), 2022, Auxiliary Machine Systems

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2022, Requirements for Preservation of Ship Structures

OTHER REFERENCES

Ketema LP, Sep 2002, DXT Direct Expansion Evaporator Installation, Operation, and Maintenance Instructions.

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 AC Tech Rep. The Contractor must provide the services of a Qualified Technical Representative who is familiar with the York 130-Ton capacity, HFC-134A Air Conditioning plants to accomplish the following on site:

- Advise on manufacturer's system information.
- Assist with disassembly, inspection, repair, modification, calibration, testing and reassembly of the equipment/system.
- Ensure compliance with manufacturer's procedures and standards during system disassembly, inspection, and reassembly as applicable.

3.1.2.1. Ensure the Tech Rep has experience with the system/equipment stated above and demonstrated on their résumé.

3.1.2.2. Submit the name and résumé of the Tech Rep to the Port Engineer prior to starting work.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000.

3.1.3.1 Protective measures - environmental. The Contractor must abide by all applicable environmental protection tasks as specified in paragraph in SFLC Std Spec 0000, Appendix A (Requirements for Environmental Protection).

3.1.3.2 Protective measures – refrigerant. The Contractor must take all necessary precautions to prevent introduction of contaminants and moisture into refrigeration system piping and components by sealing all openings.

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences). Known interferences include, but are not limited to the below-listed:

- Plant piping & fittings
- Refrigerant (134a)
- Lagging
- Work benches

3.1.5 Refrigerant handling.

3.1.5.1 Personnel qualification. The Contractor must provide the Port Engineer documentation that ensure all prime and sub contractors servicing refrigerant-based Air Conditioning and Refrigeration (AC&R) equipment hold a current Environmental Protection Agency (EPA) Technician Certification, Type IV (Universal Certification), and meet all State and local regulations and licensing requirements.

3.1.5.2 Refrigerant draining and disposal. The Contractor must ensure all vessels' Air Conditioning Plant refrigerant is drained, recovered and disposed of in accordance with all Federal, state and local environmental regulations.

3.1.6 Repair plan. The Contractor must supply a written plan of work to the Port Engineer at least 24 hours in advance of beginning work. Plan must include sequencing of work, flushing procedures, source of supply for the renewal and overhaul parts and any planned deviations from this work item.

3.1.7 Guidance. The Contractor must refer to CG Drawing 750 WMSL 516-001 for requirements, location and shipboard arrangement. The Contractor must refer to CG TP S9514-F1-MMA-010, for general & safety information, parts, operation and maintenance procedures.

3.1.8 Safety. The Contractor must familiarize themselves with and comply with applicable Warning, Caution and Attention notes invoked in CG TP S9514-F1-MMA-010, SWBS 514, Jan 2012, Air Conditioning Plant HFC-134A 130 Ton Capacity Description, Operation and Maintenance Manual.

3.1.9 Rigging. The Contractor is responsible for all rigging with the following considerations:

3.1.9.1 All existing padeyes that will be used for a lift must be pull tested to 1.5 times their Working Load Limit in accordance with SFLC Standard Spec 5000 Paragraph 3.2.7. All testing must occur in the presence of a Coast Guard Inspector.

3.1.9.2. If new padeyes are required, fabricate them in accordance with Coast Guard Drawing FL-1204-3 and SFLC Standard Spec 0740. Newly fabricated padeys must be pull tested to 2 times their Working Load Limit in accordance with SFLC Standard Spec 5000 Paragraph 3.2.7. All testing must occur in the presence of a Coast Guard Inspector.

3.1.9.3. Upon completion of testing, install a label plate in accordance with SFLC Standard Spec 5000 Paragraph B2.9.

3.1.9.4. The Contractor is responsible for moving the condenser to and from the weather deck of the ship to the AC flat. The Coast Guard will crane the condenser to and from the pier to a weather deck on the cutter.

3.1.10 Vital Loop. Contractor must keep two total AC compressor from either the #2 AC Plant or #3 AC Plant operating at all times. This will require installing blank flanges in the condenser supply and discharge piping for various plants throughout the work. The Contractor may bring down the AC plant for no more than an hour at a time to install the necessary blanks.

3.2 Operational test - initial. Prior to commencement of work, the Contractor must witness Coast Guard personnel perform an initial operational test of all items or shipboard devices to be disturbed, used, repaired, or altered, to demonstrate existing operational condition. Provide documentation of the results to the Port Engineer.

3.2.1 The Contractor must submit an operational report following startup utilizing figure 2-16 of TP S9514-F1-MMA-010. Run the plant for 1 hour with measurements taken every 5 minutes.

3.2.2 Initial operational test of the Air Conditioners can be waived if operating the AC plant will risk personnel safety and/or further damage to the system. Submit a request to the Port Engineer for review.

3.3 Refrigerant evacuation. The contractor must pump down the #1A and #1B AC circuit in accordance with Paragraph 6-1.2.1 of TP S9514-F1-MMA-010. Deviation from this procedure must be authorized by the Port Engineer. Submit documentation to the Port Engineer reporting the amount of refrigerant removed.

3.3.1 For the refrigerant from #1B, the Contractor must test the refrigerant for acid and check for Air/Non-Condensable in the recovery bottle, then dispose of the refrigerant in accordance with Federal, state and local regulations and SFLC Std Spec 0000. CG Inspector must be present to verify Pressure/Temperature Relationship on recovery cylinders after recovery cylinders have settled for a minimum of 24 hours. Provide documentation of the results to the Port Engineer.

3.3.2. The Contractor must drain the lube oil from the #1B Compressor and the Oil Separator/Reservoir and dispose of it in accordance with the local, state and Federal regulations. The Contractor must also inspect the oil removed for any evidence of contamination such as water and any solid particles. Provide documentation of the results to the Port Engineer.

3.3.3 For the refrigerant from #1A, the Contractor must test the refrigerant for acid and check for Air/Non-Condensable in the recovery bottle, then retain the refrigerant to refill the system. CG Inspector must be present to verify Pressure/Temperature Relationship on recovery cylinders after recovery cylinders have settled for a minimum of 24 hours. Provide documentation of the results to the Port Engineer.

3.4 Condenser renewal. The Contractor must renew the condenser using GFP on the #1B refrigerant circuit. Model information for the condensers is shown in Table 1.

3.4.1 The Contractor must renew all gaskets and seals disturbed during the condenser renewal condenser end bells, piping and components disturbed in accomplishing this work item using new gaskets and seals.

3.4.2. Vent the air from the condenser in accordance with Paragraph 4-5.1 of TP S9514-F1-MMA-010.

3.4.3. The Contractor must install the necessary mounting brackets for the condensers in accordance with Figure 8-10 of TP S9514-F1-MMA-010. Conduct welds and NDE in accordance with SFLC Standard Spec 0740

3.4.4 In the presence of the Coast Guard inspector, the Contractor must pressure test the condenser in accordance with Paragraph 6-7.2 of TP S9514-F1-MMA-010. upon completion of the installation. The Contractor is only required to correct leaks in joints that they disturb. Provide documentation of the results to the Port Engineer.

TABLE 1

FUNCTIONAL SERVICE	MANUFACTURER	MODEL NUMBER	NOTES
Air Conditioning Unit Condenser	KETEMA	MHX-1208A-2P	Each condenser contains 84 tubes

3.5 A/C plants flush. The Contractor must plan and execute a flush of the refrigerant circuit for the #1B AC circuit. This plan must be approved by the Port Engineer 48 hours prior to execution. Components that are being renewed or overhauled do not need to be flushed. The flush must accomplish the following tasks:

- Remove particulate contamination
- Remove rust
- Remove saltwater residue and growth
- Uses techniques and materials that are compatible with the AC system.
- Include a 10 micron filter somewhere in the flushing circuit
- Remove contamination in the evaporator tubes mechanically

3.5.1. Upon completion of the flush, the Contractor is responsible for removing the flushing media and any residual moisture from the flush.

3.5.2. If fresh water is used for the flush, then the last flush must be done with distilled water prior to drying out the system.

3.6 Borescope Inspection. The Contractor must provide a borescope inspection for the #1B refrigeration circuits after flushing that includes the following components:

- Piping leading to and from the condenser barrel, oil separator and compressor.
- Refrigerant side of evaporator

3.7 Component Renewal. The Contractor must renew the following components in Table 2 of refrigeration circuit #1B in accordance with TPS9514-F1-MMA-010. The Contractor must provide any materials that are not furnished as GFE Any brazing on the A/C unit requires the AC Tech Rep on site to oversee the nitrogen purge during brazing. Comply with all requirements of SFLC Standard Spec 0740 EXCEPT do not flush with water.

TABLE 2

Component	Dwg 077-18317-001 Item Number	Quantity	York Part Number	Manufacturer/Model
Compressor, Screw Semi-Hermetic	2	1	364-49094-215	York DXS45LA-A46/50
Separator, Oil, Right Hand	3	1	075-56077-001	Westermeyer OS6-I5R
Separator, Oil, Left Hand	4	1	075-56077-002	Westermeyer OS6-I5L
Control, High Refrigerant Pressure	9	1	025-35948-000	Alco PS3-XF5HNB297
Transducer, Suction Pressure	15	1	025-29583-000	Danfoss 060G2154
Transducer, Discharge Pressure	16	1	025-29139-001	Danfoss 060G2153
Transducer, Oil Pressure	17	1	025-29139-001	Danfoss 060G2153
Valve, Solenoid, Liquid Refrigerant Injection	26	1	025-00245-000	Danfoss 032F4023
Valve, Solenoid, Capacity Control	27	1	025-33205-000	NA
Core, Dehydrator*	30	3	026-37540-000	Emerson HX-48
Indicator, Moisture, Cartridge	42	2	026-32800-000	Sporlan AC-20
Economizer Thermal Expansion Valve.**	24	1	075-56077-002	Emerson

*The Contractor must remove all rust from the steel components of the STAS retainer assembly. See figure 2.

**Contractor must supply the packing nut wrench

3.8 Component Overhaul. The Contractor must overhaul the following components in refrigeration circuit #1B in accordance with TPS9514-F1-MMA-010:

3.8.1 Evaporator Thermal Expansion Valve (PC 23, Dwg 077-18317-001) – renew the cage assembly (PN: X9144B13B), the power head assembly (PN: XC-726MC2B) and the superheat plug (PC: X28124-1), the gasket (PN: X13455-1). The Contractor must also supply the packing nut wrench (PN: E642).

3.8.3 Main Liquid Line Solenoid Valve (PC 25, DWG 077-18317-001) – renew internal components using the Solenoid Valve Repair Kit (PN: KS30325)

3.8.4 Air Compressor Discharge Valve (PC 80, DWG 077-18317-001) – Disassemble valve and mechanically remove all internal rust.

3.9 Post-work Leak Test. The Contractor must perform a leak test of the #1A and #1B AC circuit in accordance with Paragraph 6-2.1 of TP S9514-F1-MMA-010 upon completion of all work. The Contractor is responsible for repairing any leaks from equipment that they disturbed over the course of the work. Report results via a CFR.

3.10 A/C plants. The Contractor must restore circuit #1A and #1B to original configuration. Renew the refrigerant, oil, oil filter cartridge, and all liquid line filter/drier cores in circuit 1B.

3.10.1 Prior to charging the refrigerant, the system must be pulled down to a vacuum below 500 microns for at least 5 minutes. This may be done separately or simultaneously with the vacuum requirements in TP S9514-F1-MMA-010

3.10.2 Charging Refrigerant. Upon completion of a successful leak test, the Contractor must charge refrigerant to the #1A and #1B unit in accordance with Paragraph 6-2.3 of TP S9514-F1-MMA-010 in the presence of a CG Inspector. The #1A unit will use the retained refrigerant and the #1B unit will use new refrigerant.

3.11 Operational test – post repairs. After completion of work, in the presence of the Coast Guard Inspector, the Contractor must thoroughly test and demonstrate all items or shipboard devices that have been disturbed, used, repaired, altered, or installed to be in satisfactory operating condition. Provide documentation of the results to the Port Engineer

3.11.1 During final testing of the AC units, run the #1B and #1A circuits one at a time (no other compressors operating) for an hour. The Contractor must submit an operational report following startup utilizing figure 2-16 of TP S9514-F1-MMA-010 with measurements taken every 5 minutes. Repeat the process with the #1A circuit.

3.11.2 The Contractor must install additional drier cores as needed after the initial start to ensure the system is dry and free of acids and moisture.

3.11.3 The Contractor must collect two oil samples from the #1B compressor oil after startup. One sample must be tested per C2.1.3.1 of SFLC Std Spec 5000 and one must be provided to the Port Engineer. Provide documentation of the results to the Port Engineer.

3.12 Touch-up preservation, general. The Contractor must prepare and coat all new and disturbed exterior and interior surfaces, as applicable, to match existing adjacent surfaces in accordance with SFLC Std Spec 6310, paragraph 3.1.13 (Touch-ups and minor coating repairs.)

4. NOTES

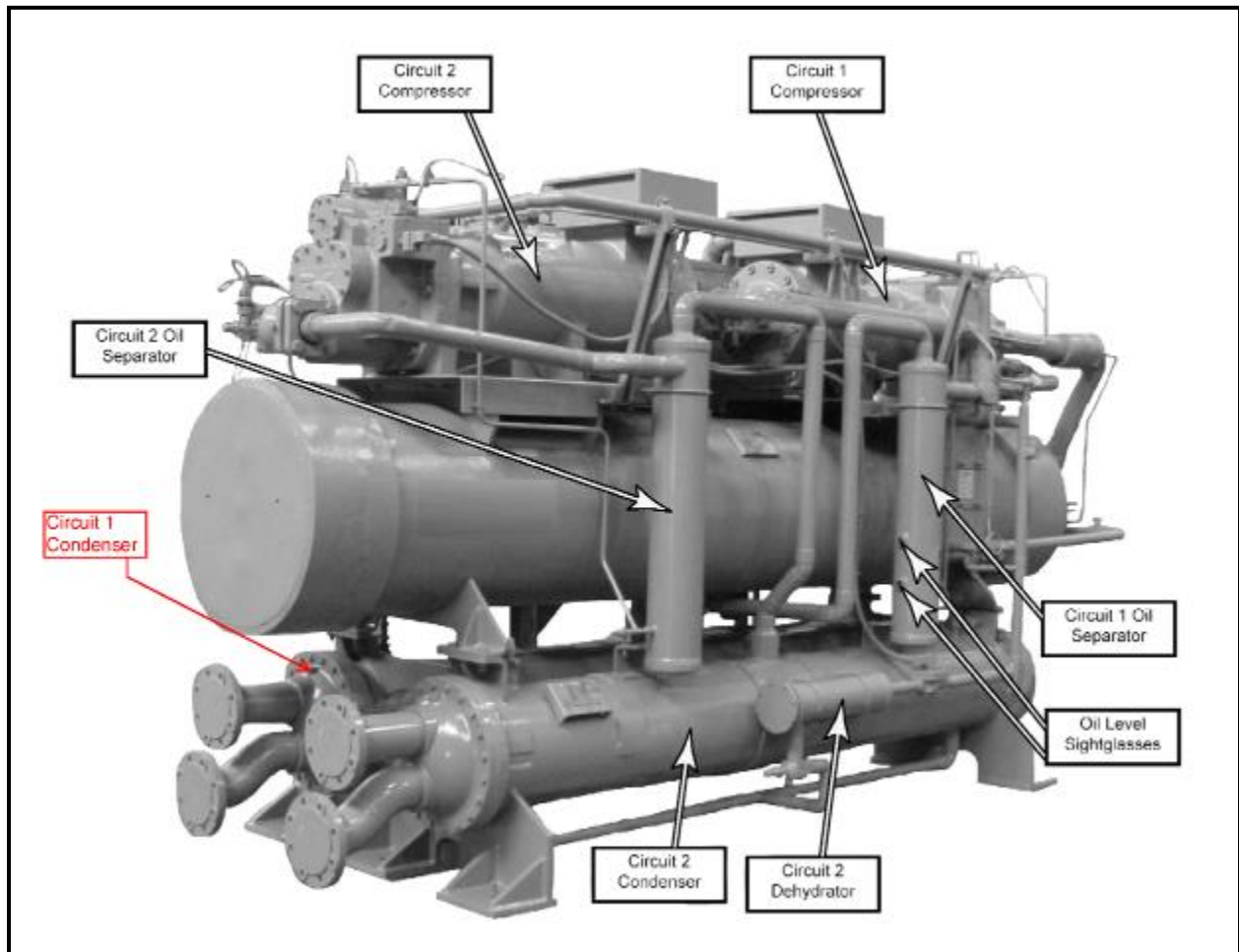


FIGURE 1. AIR CONDITIONING PLANT CONFIGURATION

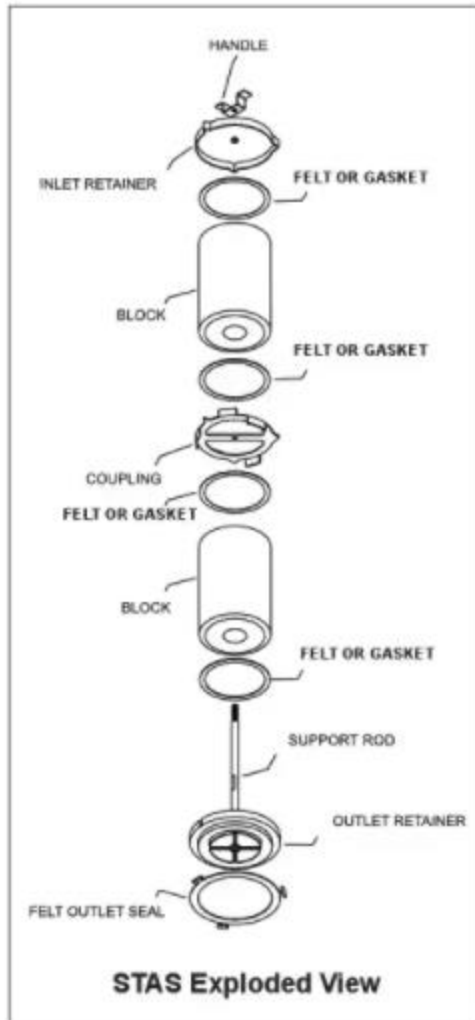


Figure 2. Filter Assembly